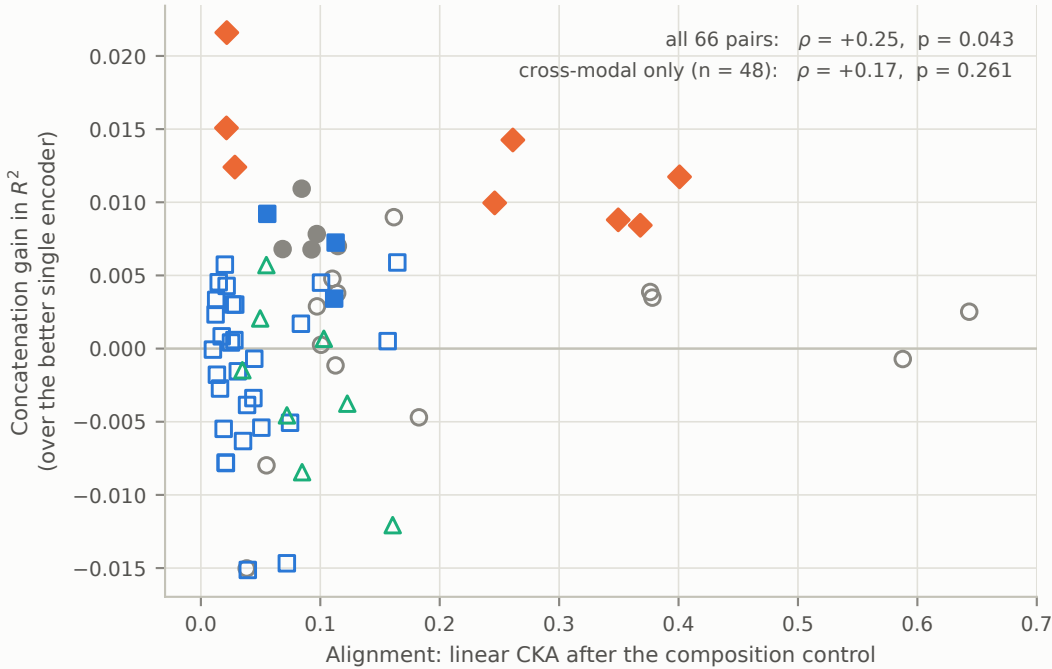
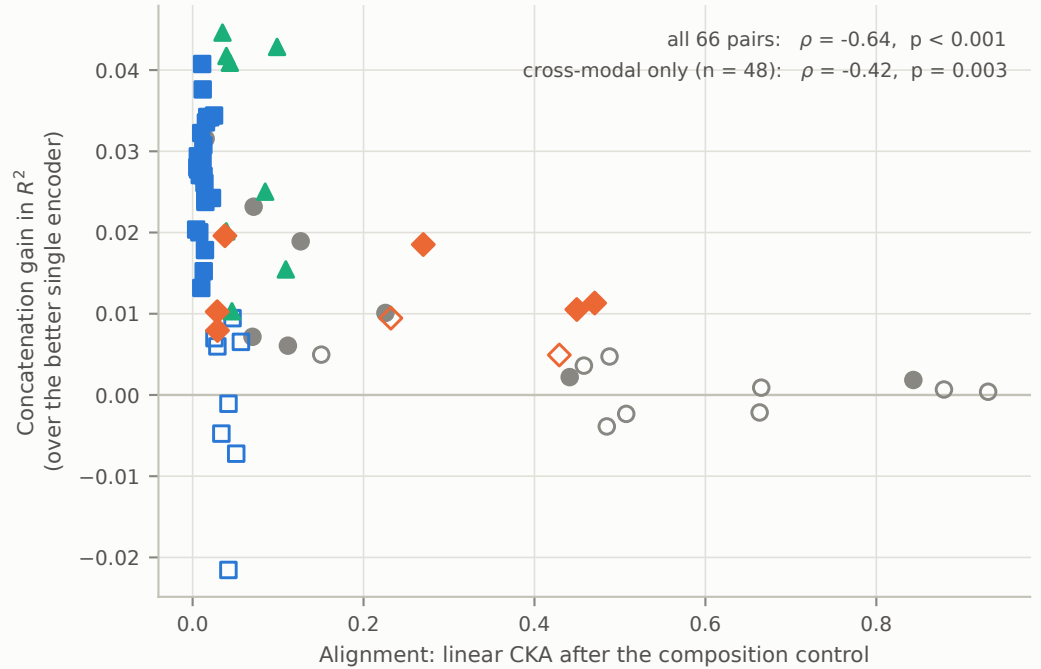


Does alignment predict where combining encoders helps?

a mRNA stability: every encoder reads one coding sequence



b GTEx: each encoder reads a different molecule



- within modality
- DNA with RNA or protein
- ▲ nucleotide RNA with protein
- ◆ codon RNA with protein
- gains under all 10 fold assignments
- gains under fewer than 10

Gain is the mean over ten further fold assignments. On GTEx the pooled correlation is carried largely by the within-modality pairs ($\rho = -0.85$, $n = 18$), where two DNA encoders read the same window and are redundant by construction; the cross-modal half is the result that bears on the Stage 2 decision. No trend line is drawn, because Spearman is a rank statistic.